JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Mathematical Reasoning

JEE Main Mathematics · Previous Year Questions with Answer Key

122	120	2008 – 2023	122	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Out of syllabus	122 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

Out of syllabus 122 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q1. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

Negation of $p \wedge (q \wedge \sim (p \wedge q))$ is :

(A) $(\sim (p \wedge q)) \wedge q$ (B) $(\sim (p \wedge q)) \vee p$ (C) $p \vee q$ (D) $\sim (p \vee q)$

Q2. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

The statement $(p \wedge (\sim q)) \vee ((\sim p) \wedge q) \vee ((\sim p) \wedge (\sim q))$ is equivalent to _____.

(A) $(\sim p) \vee (\sim q)$ (B) $p \vee (\sim q)$ (C) $\mathrm{p} \vee \mathrm{q}$ (D) $(\sim p) \vee q$

Q3. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

The negation of the statement $((A \wedge (B \vee C)) \Rightarrow (A \vee B)) \Rightarrow A$ is

(A) equivalent to $B \vee \sim C$ (B) equivalent to $\sim A$

(C) equivalent to $\sim C$ (D) a fallacy

Q4. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Among the two statements

$(\mathrm{S1}) : (p \Rightarrow q) \wedge (p \wedge (\sim q))$ is a contradiction and

$(\mathrm{S2}) : (p \wedge q) \vee ((\sim p) \wedge q) \vee (p \wedge (\sim q)) \vee ((\sim p) \wedge (\sim q))$ is a tautology

(A) both are false. (B) only (S1) is true.

(C) both are true. (D) only (S2) is true.

Q5. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

The converse of $((\sim p) \wedge q) \Rightarrow r$ is

(A) $((\sim p) \vee q) \Rightarrow r$ (B) $(\sim \mathrm{r}) \Rightarrow \mathrm{p} \wedge \mathrm{q}$

(C) $(\mathrm{p} \vee (\sim \mathrm{q})) \Rightarrow (\sim \mathrm{r})$ (D) $(\sim \mathrm{r}) \Rightarrow ((\sim \mathrm{p}) \wedge \mathrm{q})$

Q6. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

The statement $\sim [p \vee (\sim (p \wedge q))]$ is equivalent to :

(A) $(\sim (p \wedge q)) \wedge q$ (B) $\sim (p \vee q)$ (C) $(p \wedge q) \wedge (\sim p)$ (D) $\sim (p \wedge q)$

Q7. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

The negation of the statement $(p \vee q) \wedge (q \vee (\sim r))$ is :

(A) $((\sim p) \vee r)) \wedge (\sim q)$

(B) $(p \vee r) \wedge (\sim q)$

(C) $((\sim p) \vee (\sim q)) \vee (\sim r)$

(D) $((\sim p) \vee (\sim q)) \wedge (\sim r)$

Q8. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

The negation of $(p \wedge (\sim q)) \vee (\sim p)$ is equivalent to :

(A) $p \wedge q$

(B) $p \wedge (\sim q)$

(C) $p \wedge (q \wedge (\sim p))$

(D) $p \vee (q \vee (\sim p))$

Q9. JEE Main 2023 (Online) 8th April Morning Shift · MCQ (Single Correct Answer)

Negation of $(p \Rightarrow q) \Rightarrow (q \Rightarrow p)$ is :

(A) $(\sim q) \wedge p$

(B) $q \wedge (\sim p)$

(C) $p \vee (\sim q)$

(D) $(\sim p) \vee q$

Q10. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

Among the statements

(S1) : $(p \Rightarrow q) \vee ((\sim p) \wedge q)$ is a tautology

(S2) : $(q \Rightarrow p) \Rightarrow ((\sim p) \wedge q)$ is a contradiction

(A) neither (S1) and (S2) is True

(B) only (S2) is True

(C) both $(\mathrm{S1})$ and $(\mathrm{S2})$ are True

(D) only (S1) is True

Q11. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

Statement $(\mathrm{P} \Rightarrow \mathrm{Q}) \wedge (\mathrm{R} \Rightarrow \mathrm{Q})$ is logically equivalent to :

(A) $(P \Rightarrow R) \wedge (Q \Rightarrow R)$

(B) $(P \Rightarrow R) \vee (Q \Rightarrow R)$

(C) $(P \wedge R) \Rightarrow Q$

(D) $(P \vee R) \Rightarrow Q$

Q12. JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

Which of the following statements is a tautology?

(A) $\mathrm{p} \vee (\mathrm{p} \wedge \mathrm{q})$

(B) $(\mathrm{p} \wedge (\mathrm{p} \rightarrow \mathrm{q})) \rightarrow \sim \mathrm{q}$

(C) $\mathrm{p} \rightarrow (\mathrm{p} \wedge (\mathrm{p} \rightarrow \mathrm{q}))$

(D) $(\mathrm{p} \wedge \mathrm{q}) \rightarrow (\sim (\mathrm{p}) \rightarrow \mathrm{q})$

Q13. JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

The negation of the expression $q \vee ((\sim q) \wedge p)$ is equivalent to

(A) $(\sim p) \wedge (\sim q)$

(B) $(\sim p) \vee q$

(C) $p \wedge (\sim q)$

(D) $(\sim p) \vee (\sim q)$

Q14. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

The number of values of $\mathrm{r} \in \{\mathrm{p}, \mathrm{q}, \sim \mathrm{p}, \sim \mathrm{q}\}$ for which $((\mathrm{p} \wedge \mathrm{q}) \Rightarrow (\mathrm{r} \vee \mathrm{q})) \wedge ((\mathrm{p} \wedge \mathrm{r}) \Rightarrow \mathrm{q})$ is a tautology, is :

(A) 2 (B) 1 (C) 4 (D) 3

Q15. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

$(\mathrm{S1})$ $(p \Rightarrow q) \vee (p \wedge (\sim q))$ is a tautology

$(\mathrm{S2})$ $((\sim p) \Rightarrow (\sim q)) \wedge ((\sim p) \vee q)$ is a contradiction.

Then

(A) only (S2) is correct

(B) both (S1) and (S2) are correct

(C) only (S1) is correct

(D) both (S1) and (S2) are wrong

Q16. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Consider the following statements:

P : I have fever

Q: I will not take medicine

R : I will take rest.

The statement "If I have fever, then I will take medicine and I will take rest" is equivalent to :

(A) $((\sim P) \vee \sim Q) \wedge ((\sim P) \vee \sim R)$

(B) $(P \vee \sim Q) \wedge (P \vee \sim R)$

(C) $((\sim P) \vee \sim Q) \wedge ((\sim P) \vee R)$

(D) $(P \vee Q) \wedge ((\sim P) \vee R)$

Q17. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

Among the statements :

$(\mathrm{S1})$ $((\mathrm{p} \vee \mathrm{q}) \Rightarrow \mathrm{r}) \Leftrightarrow (\mathrm{p} \Rightarrow \mathrm{r})$

$(\mathrm{S2})$ $((\mathrm{p} \vee \mathrm{q}) \Rightarrow \mathrm{r}) \Leftrightarrow ((\mathrm{p} \Rightarrow \mathrm{r}) \vee (\mathrm{q} \Rightarrow \mathrm{r}))$

(A) only (S1) is a tautology

(B) neither (S1) nor (S2) is a tautology

(C) both (S1) and (S2) are tautologies

(D) only (S2) is a tautology

Q18. JEE Main 2023 (Online) 29th January Morning Shift · MCQ (Single Correct Answer)

If p, q and r are three propositions, then which of the following combination of truth values of p, q and r makes the logical expression $\{(p \vee q) \wedge ((\sim p) \vee r)\} \rightarrow ((\sim q) \vee r)$ false?

(A) $p = F, q = T, r = F$

(B) $p = T, q = T, r = F$

(C) $p = T, q = F, r = T$

(D) $p = T, q = F, r = F$

Q19. JEE Main 2023 (Online) 25th January Evening Shift · MCQ (Single Correct Answer)

Let $\Delta, \nabla \in \{\wedge, \vee\}$ be such that $(\mathrm{p} \rightarrow \mathrm{q}) \Delta (\mathrm{p} \nabla \mathrm{q})$ is a tautology. Then

(A) $\Delta = \vee, \nabla = \vee$

(B) $\Delta = \vee, \nabla = \wedge$

(C) $\Delta = \wedge, \nabla = \wedge$

(D) $\Delta = \wedge, \nabla = \vee$

Q20. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

The statement $(p \wedge (\sim q)) \Rightarrow (p \Rightarrow (\sim q))$ is

(A) a tautology

(B) equivalent to $(\sim p) \vee (\sim q)$

(C) a contradiction

(D) $p \vee q$

Q21. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

Let p and q be two statements. Then $\sim (p \wedge (p \Rightarrow \sim q))$ is equivalent to

(A) $(\sim p) \vee q$

(B) $p \vee (p \wedge (\sim q))$

(C) $p \vee (p \wedge q)$

(D) $p \vee ((\sim p) \wedge q)$

Q22. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

The compound statement $(\sim (P \wedge Q)) \vee ((\sim P) \wedge Q) \Rightarrow ((\sim P) \wedge (\sim Q))$ is equivalent to

(A) $((\sim P) \vee Q) \wedge (\sim Q)$

(B) $(\sim Q) \vee P$

(C) $((\sim P) \vee Q) \wedge ((\sim Q) \vee P)$

(D) $(\sim P) \vee Q$

Q23. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

The statement $(p \Rightarrow q) \vee (p \Rightarrow r)$ is NOT equivalent to

(A) $(p \wedge (\sim r)) \Rightarrow q$

(B) $(\sim q) \Rightarrow ((\sim r) \vee p)$

(C) $p \Rightarrow (q \vee r)$

(D) $(p \wedge (\sim q)) \Rightarrow r$

Q24. JEE Main 2022 (Online) 29th July Morning Shift · MCQ (Single Correct Answer)

The statement $(p \wedge q) \Rightarrow (p \wedge r)$ is equivalent to :

(A) $q \Rightarrow (p \wedge r)$

(B) $p \Rightarrow (\mathrm{p} \wedge \mathrm{r})$

(C) $(\mathrm{p} \wedge \mathrm{r}) \Rightarrow (\mathrm{p} \wedge \mathrm{q})$

(D) $(p \wedge q) \Rightarrow r$

Q25. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

Let

p : Ramesh listens to music.

q : Ramesh is out of his village.

r : It is Sunday.

s : It is Saturday.

Then the statement "Ramesh listens to music only if he is in his village and it is Sunday or Saturday" can be expressed as

(A) $((\sim q) \wedge (r \vee s)) \Rightarrow p$

(B) $(\mathrm{q} \wedge (\mathrm{r} \vee \mathrm{s})) \Rightarrow \mathrm{p}$

(C) $p \Rightarrow (q \wedge (r \vee s))$

(D) $\mathrm{p} \Rightarrow ((\sim \mathrm{q}) \wedge (\mathrm{r} \vee \mathrm{s}))$

Q26. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

Let the operations $*, \odot \in \{\wedge, \vee\}$. If $(\mathrm{p} * \mathrm{q}) \odot (\mathrm{p} \odot \sim \mathrm{q})$ is a tautology, then the ordered pair $(*, \odot)$ is :

(A) $(\vee, \wedge)$

(B) $(\vee, \vee)$

(C) $(\wedge, \wedge)$

(D) $(\wedge, \vee)$

Q27. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

If the truth value of the statement $(P \wedge (\sim R)) \rightarrow ((\sim R) \wedge Q)$ is F, then the truth value of which of the following is F ?

(A) $\mathrm{P} \vee \mathrm{Q} \rightarrow \sim \mathrm{R}$

(B) $\mathrm{R} \vee \mathrm{Q} \rightarrow \sim \mathrm{P}$

(C) $\sim (\mathrm{P} \vee \mathrm{Q}) \rightarrow \sim \mathrm{R}$

(D) $\sim (\mathrm{R} \vee \mathrm{Q}) \rightarrow \sim \mathrm{P}$

Q28. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

$(p \wedge r) \Leftrightarrow (p \wedge (\sim q))$ is equivalent to $(\sim p)$ when r is

(A) p

(B) $\sim p$

(C) q

(D) $\sim q$

Q29. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

Negation of the Boolean expression $p \Leftrightarrow (q \Rightarrow p)$ is

(A) $(\sim p) \wedge q$

(B) $p \wedge (\sim q)$

(C) $(\sim p) \vee (\sim q)$

(D) $(\sim p) \wedge (\sim q)$

Q30. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

The statement $(\sim (\mathrm{p} \Leftrightarrow \sim \mathrm{q})) \wedge \mathrm{q}$ is :

(A) a tautology

(B) a contradiction

(C) equivalent to $(p \Rightarrow q) \wedge q$

(D) equivalent to $(p \Rightarrow q) \wedge p$

Q31. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Consider the following statements:

P : Ramu is intelligent.

Q : Ramu is rich.

R : Ramu is not honest.

The negation of the statement "Ramu is intelligent and honest if and only if Ramu is not rich" can be expressed as:

(A) $((P \wedge (\sim R)) \wedge Q) \wedge ((\sim Q) \wedge ((\sim P) \vee R))$

(B) $((P \wedge R) \wedge Q) \vee ((\sim Q) \wedge ((\sim P) \vee (\sim R)))$

(C) $((P \wedge R) \wedge Q) \wedge ((\sim Q) \wedge ((\sim P) \vee (\sim R)))$

(D) $((P \wedge (\sim R)) \wedge Q) \vee ((\sim Q) \wedge ((\sim P) \vee R))$

Q32. JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Which of the following statements is a tautology ?

(A) $((\sim \mathrm{p}) \vee \mathrm{q}) \Rightarrow \mathrm{p}$

(B) $p \Rightarrow ((\sim p) \vee q)$

(C) $((\sim p) \vee q) \Rightarrow q$

(D) $q \Rightarrow ((\sim p) \vee q)$

Q33. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

The conditional statement

$((p \wedge q) \rightarrow ((\sim p) \vee r)) \vee (((\sim p) \vee r) \rightarrow (p \wedge q))$ is :

(A) a tautology

(B) a contradiction

(C) equivalent to $p \wedge q$

(D) equivalent to $(\sim p) \vee r$

Q34. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Negation of the Boolean statement (p $\vee$ q) $\Rightarrow$ (($\sim$ r) $\vee$ p) is equivalent to :

(A) p $\wedge$ ($\sim$ q) $\wedge$ r

(B) ($\sim$ p) $\wedge$ ($\sim$ q) $\wedge$ r

(C) ($\sim$ p) $\wedge$ q $\wedge$ r

(D) p $\wedge$ q $\wedge$ ($\sim$ r)

Q35. JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

Let $\Delta \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow\}$ be such that $(p \wedge q) \Delta ((p \vee q) \Rightarrow q)$ is a tautology. Then Δ is equal to :

(A) $\wedge$

(B) $\vee$

(C) $\Rightarrow$

(D) $\Leftrightarrow$

Q36. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Let p, q, r be three logical statements. Consider the compound statements

$S_1 : ((\sim p) \vee q) \vee ((\sim p) \vee r)$ and

$S_2 : p \rightarrow (q \vee r)$

Then, which of the following is NOT true?

(A) If S_2 is True, then S_1 is True

(B) If S_2 is False, then S_1 is False

(C) If S_2 is False, then S_1 is True

(D) If S_1 is False, then S_2 is False

Q37. JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Which of the following statement is a tautology?

(A) $((\sim q) \wedge p) \wedge q$

(B) $((\sim q) \wedge p) \wedge (p \wedge (\sim p))$

(C) $((\sim q) \wedge p) \vee (p \vee (\sim p))$

(D) $(p \wedge q) \wedge (\sim p \wedge q))$

Q38. JEE Main 2022 (Online) 27th June Morning Shift · MCQ (Single Correct Answer)

The boolean expression $(\sim (p \wedge q)) \vee q$ is equivalent to :

(A) $q \rightarrow (p \wedge q)$

(B) $p \rightarrow q$

(C) $p \rightarrow (p \rightarrow q)$

(D) $p \rightarrow (p \vee q)$

Q39. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

Let r $\in$ {p, q, $\sim$p, $\sim$q} be such that the logical statement

r $\vee$ ($\sim$p) $\Rightarrow$ (p $\wedge$ q) $\vee$ r

is a tautology. Then r is equal to :

(A) p

(B) q

(C) $\sim$p

(D) $\sim$q

Q40. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let $\Delta, \nabla \in \{\wedge, \vee\}$ be such that p ∇ q $\Rightarrow$ ((p Δ q) ∇ r) is a tautology. Then (p ∇ q) Δ r is logically equivalent to :

(A) (p Δ r) $\vee$ q

(B) (p Δ r) $\wedge$ q

(C) (p $\wedge$ r) Δ q

(D) (p ∇ r) $\wedge$ q

Q41. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

The negation of the Boolean expression $((\sim q) \wedge p) \Rightarrow ((\sim p) \vee q)$ is logically equivalent to :

(A) $p \Rightarrow q$ (B) $q \Rightarrow p$ (C) $\sim (p \Rightarrow q)$ (D) $\sim (q \Rightarrow p)$

Q42. JEE Main 2022 (Online) 25th June Morning Shift · MCQ (Single Correct Answer)

Consider the following two propositions:

$P1 : \sim (p \rightarrow \sim q)$

$P2 : (p \wedge \sim q) \wedge ((\sim p) \vee q)$

If the proposition $p \rightarrow ((\sim p) \vee q)$ is evaluated as FALSE, then :

(A) P1 is TRUE and P2 is FALSE (B) P1 is FALSE and P2 is TRUE

(C) Both P1 and P2 are FALSE (D) Both P1 and P2 are TRUE

Q43. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

Consider the following statements:

A : Rishi is a judge.

B : Rishi is honest.

C : Rishi is not arrogant.

The negation of the statement "if Rishi is a judge and he is not arrogant, then he is honest" is

(A) $B \rightarrow (A \vee C)$ (B) $(\sim B) \wedge (A \wedge C)$

(C) $B \rightarrow ((\sim A) \vee (\sim C))$ (D) $B \rightarrow (A \wedge C)$

Q44. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

The number of choices for $\Delta \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow\}$, such that

$(p \Delta q) \Rightarrow ((p \Delta \sim q) \vee ((\sim p) \Delta q))$ is a tautology, is :

(A) 1 (B) 2 (C) 3 (D) 4

Q45. JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

Which of the following is equivalent to the Boolean expression $p \wedge \sim q$?

(A) $\sim (q \rightarrow p)$ (B) $\sim p \rightarrow \sim q$ (C) $\sim (p \rightarrow \sim q)$ (D) $\sim (p \rightarrow q)$

Q46. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

Negation of the statement $(p \vee r) \Rightarrow (q \vee r)$ is :

(A) $p \wedge \sim q \wedge \sim r$ (B) $\sim p \wedge q \wedge \sim r$

(C) $\sim p \wedge q \wedge r$ (D) $p \wedge q \wedge r$

Q47. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

Let *, $\square \in\{\wedge, \vee\}$ be such that the Boolean expression $(p * \sim q) \Rightarrow (p \square q)$ is a tautology. Then :

(A) $* = \vee, \square = \vee$

(B) $* = \wedge, \square = \wedge$

(C) $* = \wedge, \square = \vee$

(D) $* = \vee, \square = \wedge$

Q48. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

The Boolean expression $(p \wedge q) \Rightarrow ((r \wedge q) \wedge p)$ is equivalent to :

(A) $(p \wedge q) \Rightarrow (r \wedge q)$

(B) $(q \wedge r) \Rightarrow (p \wedge q)$

(C) $(p \wedge q) \Rightarrow (r \vee q)$

(D) $(p \wedge r) \Rightarrow (p \wedge q)$

Q49. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

The statement $(p \wedge (p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow r$ is :

(A) a tautology

(B) equivalent to $p \rightarrow \sim r$

(C) a fallacy

(D) equivalent to $q \rightarrow \sim r$

Q50. JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

Consider the two statements :

(S1) : $(p \rightarrow q) \vee (\sim q \rightarrow p)$ is a tautology .

(S2) : $(p \wedge \sim q) \wedge (\sim p \vee q)$ is a fallacy.

Then :

(A) only (S1) is true.

(B) both (S1) and (S2) are false.

(C) both (S1) and (S2) are true.

(D) only (S2) is true.

Q51. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

If the truth value of the Boolean expression $((p \vee q) \wedge (q \rightarrow r) \wedge (\sim r)) \rightarrow (p \wedge q)$ is false, then the truth values of the statements p, q, r respectively can be :

(A) T F T

(B) F F T

(C) T F F

(D) F T F

Q52. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Which of the following is the negation of the statement "for all M > 0, there exists x∈S such that x $\geq$ M" ?

(A) there exists M > 0, such that x < M for all x∈S

(B) there exists M > 0, there exists x∈S such that x $\geq$ M

(C) there exists M > 0, there exists x∈S such that x < M

(D) there exists M > 0, such that x $\geq$ M for all x∈S

Q53. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

The compound statement $(P \vee Q) \wedge (\sim P) \Rightarrow Q$ is equivalent to :

(A) $P \vee Q$

(B) $P \wedge \sim Q$

(C) $\sim (P \Rightarrow Q)$

(D) $\sim (P \Rightarrow Q) \Leftrightarrow P \wedge \sim Q$

Q54. JEE Main 2021 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Consider the statement "The match will be played only if the weather is good and ground is not wet". Select the correct negation from the following :

(A) The match will not be played and weather is not good and ground is wet.

(B) If the match will not be played, then either weather is not good or ground is wet.

(C) The match will be played and weather is not good or ground is wet.

(D) The match will not be played or weather is good and ground is not wet.

Q55. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

The Boolean expression $(p \Rightarrow q) \wedge (q \Rightarrow \sim p)$ is equivalent to :

(A) $\sim$ q

(B) q

(C) p

(D) $\sim$ p

Q56. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Which of the following Boolean expressions is not a tautology?

(A) (p $\Rightarrow$ q) $\vee$ ($\sim$ q $\Rightarrow$ p)

(B) (q $\Rightarrow$ p) $\vee$ ($\sim$ q $\Rightarrow$ p)

(C) (p $\Rightarrow$ $\sim$ q) $\vee$ ($\sim$ q $\Rightarrow$ p)

(D) ($\sim$ p $\Rightarrow$ q) $\vee$ ($\sim$ q $\Rightarrow$ p)

Q57. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

Consider the following three statements :

(A) If 3 + 3 = 7 then 4 + 3 = 8

(B) If 5 + 3 = 8 then earth is flat.

(C) If both (A) and (B) are true then 5 + 6 = 17.

Then, which of the following statements is correct?

(A) (A) is false, but (B) and (C) re true

(B) (A) and (C) are true while (B) is false

(C) (A) is true while (B) and (C) are false

(D) (A) and (B) are false while (C) is true

Q58. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

The Boolean expression $(p \wedge \sim q) \Rightarrow (q \vee \sim p)$ is equivalent to :

(A) $q \Rightarrow p$

(B) $p \Rightarrow q$

(C) $\sim q \Rightarrow p$

(D) $p \Rightarrow \sim q$

Q59. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

If P and Q are two statements, then which of the following compound statements is a tautology?

(A) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow (P \wedge Q)$

(B) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow Q$

(C) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow P$

(D) $((P \Rightarrow Q) \wedge \sim Q) \Rightarrow \sim P$

Q60. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

If the Boolean expression $(p \wedge q) \odot (p \otimes q)$ is a tautology, then $\odot$ and $\otimes$ are respectively given by :

(A) $\vee, \rightarrow$

(B) $\rightarrow, \rightarrow$

(C) $\wedge, \vee$

(D) $\wedge, \rightarrow$

Q61. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

If the Boolean expression $(p \Rightarrow q) \Leftrightarrow (q * (\sim p)$ is a tautology, then the boolean expression $(p * (\sim q))$ is equivalent to :

(A) $q \Rightarrow p$

(B) $p \Rightarrow q$

(C) $p \Rightarrow \sim q$

(D) $\sim q \Rightarrow p$

Q62. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

Which of the following Boolean expression is a tautology?

(A) $(p \wedge q) \vee (p \rightarrow q)$

(B) $(p \wedge q) \vee (p \vee q)$

(C) $(p \wedge q) \rightarrow (p \rightarrow q)$

(D) $(p \wedge q) \wedge (p \rightarrow q)$

Q63. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

Let $F_1(A, B, C) = (A \wedge \sim B) \vee [\sim C \wedge (A \vee B)] \vee \sim A$ and

$F_2(A, B) = (A \vee B) \vee (B \rightarrow \sim A)$ be two logical expressions. Then :

(A) Both F_1 and F_2 are not tautologies

(B) F_1 and F_2 both are tautologies

(C) F_1 is not a tautology but F_2 is a tautology

(D) F_1 is a tautology but F_2 is not a tautology

Q64. JEE Main 2021 (Online) 25th February Evening Shift · MCQ (Single Correct Answer)

The contrapositive of the statement "If you will work, you will earn money" is :

(A) If you will not earn money, you will not work

(B) If you will earn money, you will work

(C) You will earn money, if you will not work

(D) To earn money, you need to work

Q65. JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

The statement $A \rightarrow (B \rightarrow A)$ is equivalent to :

(A) $A \rightarrow (A \longleftrightarrow B)$

(B) $A \rightarrow (A \vee B)$

(C) $A \rightarrow (A \wedge B)$

(D) $A \rightarrow (A \rightarrow B)$

Q66. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

The negation of the statement

$\sim p \wedge (p \vee q)$ is :

(A) $p \vee \sim q$

(B) $\sim p \vee q$

(C) $\sim p \wedge q$

(D) $p \wedge \sim q$

Q67. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

For the statements p and q, consider the following compound statements :

(a) $(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$

(b) $((p \vee q) \wedge \sim p) \rightarrow q$

Then which of the following statements is correct?

(A) (b) is a tautology but not (a).

(B) (a) and (b) both are not tautologies.

(C) (a) and (b) both are tautologies.

(D) (a) is a tautology but not (b).

Q68. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

The statement among the following that is a tautology is :

(A) $B \rightarrow [A \wedge (A \rightarrow B)]$

(B) $[A \wedge (A \rightarrow B)] \rightarrow B$

(C) $[A \wedge (A \vee B)]$

(D) $[A \vee (A \wedge B)]$

Q69. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

Consider the statement :

"For an integer n, if $n^3 - 1$ is even, then n is odd."

The contrapositive statement of this statement is :

(A) For an integer n, if n is even, then $n^3 - 1$ is even.

(B) For an integer n, if $n^3 - 1$ is not even, then n is not odd.

(C) For an integer n, if n is odd, then $n^3 - 1$ is even.

(D) For an integer n, if n is even, then $n^3 - 1$ is odd.

Q70. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

The negation of the Boolean expression $p \vee (\sim p \wedge q)$ is equivalent to :

(A) $p \wedge \sim q$

(B) $\sim p \vee \sim q$

(C) $\sim p \wedge q$

(D) $\sim p \wedge \sim q$

Q71. JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

The statement

$(p \rightarrow (q \rightarrow p)) \rightarrow (p \rightarrow (p \vee q))$ is :

(A) a tautology

(B) a contradiction

(C) equivalent to $(p \vee q) \wedge (\sim p)$

(D) equivalent to $(p \wedge q) \vee (\sim q)$

Q72. JEE Main 2020 (Online) 5th September Morning Slot · MCQ (Single Correct Answer)

The negation of the Boolean expression x $\leftrightarrow$ ~ y is equivalent to :

(A) $(x \wedge y) \vee (\sim x \wedge \sim y)$

(B) $(\sim x \wedge y) \vee (\sim x \wedge \sim y)$

(C) $(x \wedge y) \wedge (\sim x \vee \sim y)$

(D) $(x \wedge \sim y) \vee (\sim x \wedge y)$

Q73. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

Contrapositive of the statement :
'If a function f is differentiable at a, then it is also continuous at a', is:

(A) If a function f is continuous at a, then it is not differentiable at a.

(B) If a function f is not continuous at a, then it is differentiable at a.

(C) If a function f is not continuous at a, then it is not differentiable at a.

(D) If a function f is continuous at a, then it is differentiable at a.

Q74. JEE Main 2020 (Online) 4th September Morning Slot · MCQ (Single Correct Answer)

Given the following two statements:

$(S_1) : (q \vee p) \rightarrow (p \leftrightarrow \sim q)$ is a tautology

$(S_2) : \sim q \wedge (\sim p \leftrightarrow q)$ is a fallacy. Then:

(A) both (S_1) and (S_2) are not correct

(B) only (S_1) is correct

(C) only (S_2) is correct

(D) both (S_1) and (S_2) are correct

Q75. JEE Main 2020 (Online) 3rd September Evening Slot · MCQ (Single Correct Answer)

Let p, q, r be three statements such that the truth value of
$(p \wedge q) \rightarrow (\sim q \vee r)$ is F. Then the truth values of p, q, r are respectively :

(A) T, F, T

(B) F, T, F

(C) T, T, T

(D) T, T, F

Q76. JEE Main 2020 (Online) 3rd September Morning Slot · MCQ (Single Correct Answer)

The proposition p $\rightarrow$ ~ (p $\wedge$ ~q) is equivalent to :

(A) $(\sim p) \vee q$

(B) q

(C) $(\sim p) \wedge q$

(D) $(\sim p) \vee (\sim q)$

Q77. JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

Which of the following is a tautology ?

(A) $(\sim p) \wedge (p \vee q) \rightarrow q$

(B) $(q \rightarrow p) \vee \sim (p \rightarrow q)$

(C) $(p \rightarrow q) \wedge (q \rightarrow p)$

(D) $(\sim q) \vee (p \wedge q) \rightarrow q$

Q78. JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

The contrapositive of the statement
"If I reach the station in time, then I will catch the train" is :

(A) If I will catch the train, then I reach the station in time.

(B) If I do not reach the station in time, then I will not catch the train.

(C) If I will not catch the train, then I do not reach the station in time.

(D) If I do not reach the station in time, then I will catch the train.

Q79. JEE Main 2020 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

If $p \rightarrow (p \wedge \sim q)$ is false, then the truth values of p and q are respectively :

(A) T, T (B) T, F (C) F, T (D) F, F

Q80. JEE Main 2020 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Negation of the statement :

$\sqrt{5}$ is an integer or 5 is irrational is :

(A) $\sqrt{5}$ is not an integer and 5 is not irrational.

(B) $\sqrt{5}$ is irrational or 5 is an integer.

(C) $\sqrt{5}$ is an integer and 5 is irrational.

(D) $\sqrt{5}$ is not an integer or 5 is not irrational.

Q81. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

Which of the following statements is a tautology?

(A) $\sim(p \wedge \sim q) \rightarrow p \vee q$ (B) $\sim(p \vee \sim q) \rightarrow p \vee q$

(C) $\sim(p \vee \sim q) \rightarrow p \wedge q$ (D) $p \vee (\sim q) \rightarrow p \wedge q$

Q82. JEE Main 2020 (Online) 8th January Morning Slot · MCQ (Single Correct Answer)

Which one of the following is a tautology?

(A) $P \wedge (P \vee Q)$ (B) $P \vee (P \wedge Q)$

(C) $Q \rightarrow (P \wedge (P \rightarrow Q))$ (D) $(P \wedge (P \rightarrow Q)) \rightarrow Q$

Q83. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

Let A, B, C and D be four non-empty sets. The contrapositive statement of "If $A \subseteq B$ and $B \subseteq D$, then $A \subseteq C$" is :

(A) If $A \not\subseteq C$, then $A \not\subseteq B$ or $B \not\subseteq D$

(B) If $A \not\subseteq C$, then $A \not\subseteq B$ and $B \subseteq D$

(C) If $A \subseteq C$, then $B \subset A$ or $D \subset B$

(D) If $A \not\subseteq C$, then $A \subseteq B$ and $B \subseteq D$

Q84. JEE Main 2020 (Online) 7th January Morning Slot · MCQ (Single Correct Answer)

The logical statement $(p \Rightarrow q) \Lambda (q \Rightarrow \sim p)$ is equivalent to :

(A) q (B) $\sim p$ (C) p (D) $\sim q$

Q85. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

The Boolean expression $\sim(p \Rightarrow (\sim q))$ is equivalent to :

(A) $p \wedge q$ (B) $q \Rightarrow \sim p$ (C) $p \vee q$ (D) $(\sim p) \Rightarrow q$

Q86. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

If the truth value of the statement $p \rightarrow (\sim q \vee r)$ is false (F), then the truth values of the statements p, q, r are respectively :

(A) T, F, T (B) F, T, T (C) T, T, F (D) T, F, F

Q87. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

The negation of the Boolean expression $\sim s \vee (\sim r \wedge s)$ is equivalent to :

(A) $\sim s \wedge \sim r$ (B) r (C) $s \vee r$ (D) $s \wedge r$

Q88. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Which one of the following Boolean expressions is a tautology?

(A) $(p \vee q) \wedge (\sim p \vee \sim q)$ (B) $(p \vee q) \vee (p \vee \sim q)$

(C) $(p \wedge q) \vee (p \wedge \sim q)$ (D) $(p \vee q) \wedge (p \vee \sim q)$

Q89. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

If $p \Rightarrow (q \vee r)$ is false, then the truth values of p, q, r are respectively :-

(A) F, F, F (B) T, F, F (C) F, T, T (D) T, T, F

Q90. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

For any two statements p and q, the negation of the expression $p \vee (\sim p \wedge q)$ is :

(A) $p \leftrightarrow q$ (B) $\sim p \wedge \sim q$ (C) $p \wedge q$ (D) $\sim p \vee \sim q$

Q91. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

Which one of the following statements is not a tautology?

(A) $(p \wedge q) \rightarrow (\sim p) \vee q$ (B) $(p \wedge q) \rightarrow p$

(C) $(p \vee q) \rightarrow (p \vee (\sim q))$ (D) $p \rightarrow (p \vee q)$

Q92. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The contrapositive of the statement "If you are born in India, then you are a citizen of India", is :

(A) If you are not a citizen of India, then you are not born in India.

(B) If you are born in India, then you are not a citizen of India.

(C) If you are a citizen of India, then you are born in India.

(D) If you are not born in India, then you are not a citizen of India

Q93. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

The expression $\sim (\sim p \rightarrow q)$ is logically equivalent to :

(A) $p \wedge q$ (B) $\sim p \wedge \sim q$ (C) $p \wedge \sim q$ (D) $\sim p \wedge q$

Q94. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

The Boolean expression $((p \wedge q) \vee (p \vee \sim q)) \wedge (\sim p \wedge \sim q)$ is equivalent to :

(A) $p \wedge q$ (B) $p \wedge (\sim q)$

(C) $p \vee (\sim q)$ (D) $(\sim p) \wedge (\sim q)$

Q95. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Contrapositive of the statement " If two numbers are not equal, then their squares are not equal." is :

(A) If the squares of two numbers are equal, then the numbers are not equal

(B) If the squares of two numbers are equal, then the numbers are equal

(C) If the squares of two numbers are not equal, then the numbers are equal

(D) If the squares of two numbers are not equal, then the numbers are not equal

Q96. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

If q is false and $p \wedge q \leftrightarrow r$ is true, then which one of the following statements is a tautology ?

(A) $p \wedge r$

(B) $(p \vee r) \rightarrow (p \wedge r)$

(C) $p \vee r$

(D) $(p \wedge r) \rightarrow (p \vee r)$

Q97. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

Consider the following three statements :

P : 5 is a prime number

Q : 7 is a factor of 192

R : L.C.M. of 5 and 7 is 35

Then the truth value of which one of the following statements is true ?

(A) $(P \wedge Q) \vee (\sim R)$

(B) $P \vee (\sim Q \wedge R)$

(C) $(\sim P) \wedge (\sim Q \wedge R)$

(D) $(\sim P) \vee (Q \wedge R)$

Q98. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

Consider the statement : "P(n) : $n^2 - n + 41$ is prime". Then which one of the following is true ?

(A) P(5) is false but P(3) is true

(B) Both P(3) and P(5) are true

(C) P(3) is false but P(5) is true

(D) Both P(3) and P(5) are false

Q99. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

The logical statement

$[\sim(\sim p \vee q) \vee (p \wedge r)] \wedge (\sim q \wedge r)$ is equivalent to :

(A) $(\sim p \wedge \sim q) \wedge r$

(B) $\sim p \vee r$

(C) $(p \wedge r) \wedge \sim q$

(D) $(p \wedge \sim q) \vee r$

Q100. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

If the Boolean expression
$(p \oplus q) \wedge (\sim p \odot q)$ is equivalent
to $p \wedge q$, where $\oplus, \odot \in \{\wedge, \vee\}$, then the
ordered pair $(\oplus, \odot)$ is :

(A) $(\vee, \wedge)$

(B) $(\vee, \vee)$

(C) $(\wedge, \vee)$

(D) $(\wedge, \wedge)$

Q101. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

If $p \rightarrow (\sim p \vee \sim q)$ is false, then the truth values of p and q are respectively :

(A) F, F (B) T, F (C) F, T (D) T, T

Q102. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

The Boolean expression

$\sim (p \vee q) \vee (\sim p \wedge q)$ is equvalent to :

(A) $\sim q$ (B) $\sim p$ (C) p (D) q

Q103. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

Consider the following two statements :

Statement p :

The value of sin 120° can be derived by taking $\theta = 240^o$ in the equation
$2\sin\frac{\theta}{2} = \sqrt{1+\sin\theta} - \sqrt{1-\sin\theta}$

Statement q :

The angles A, B, C and D of any quadrilateral ABCD satisfy the equation
$\cos\left(\frac{1}{2}(A+C)\right) + \cos\left(\frac{1}{2}(B+D)\right) = 0$

Then the truth values of p and q are respectively :

(A) F, T (B) T, F (C) T, T (D) F, F

Q104. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

If $(p \wedge \sim q) \wedge (p \wedge r) \rightarrow \sim p \vee q$ is false, then the truth values of p, q and r are, respectively :

(A) F, T, F (B) T, F, T (C) T, T, T (D) F, F, F

Q105. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Contrapositive of the statement

'If two numbers are not equal, then their squares are not equal', is :

(A) If the squares of two numbers are equal, then the numbers are equal.

(B) If the squares of two numbers are equal, then the numbers are not equal.

(C) If the squares of two numbers are not equal, then the numbers are not equal.

(D) If the squares of two numbers are not equal, then the numbers are equal.

Q106. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The proposition $(\sim p) \vee (p \wedge \sim q)$ is equivalent to :

(A) p $\vee$ ~ q (B) p $\rightarrow$ ~ q (C) p $\wedge$ ~ q (D) q $\rightarrow$ p

Q107. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The following statement

$(p \rightarrow q) \rightarrow [(\sim p \rightarrow q) \rightarrow q]$ is :

(A) equivalent to $\sim p \rightarrow q$

(B) equivalent to $p \rightarrow \sim q$

(C) a fallacy

(D) a tautology

Q108. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The contrapositive of the following statement,

"If the side of a square doubles, then its area increases four times", is :

(A) If the side of a square is not doubled, then its area does not increase four times.

(B) If the area of a square increases four times, then its side is doubled.

(C) If the area of a square increases four times, then its side is not doubled.

(D) If the area of a square does not increase four times, then its side is not doubled.

Q109. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Consider the following two statements :

P : If 7 is an odd number, then 7 is divisible by 2.
Q : If 7 is a prime number, then 7 is an odd number

If V_1 is the truth value of the contrapositive of P and V_2 is the truth value of contrapositive of Q, then the ordered pair (V_1 , V_2) equals :

(A) (T, T) (B) (T, F) (C) (F, T) (D) (F, F)

Q110. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

The Boolean expression

$(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$ is equivalent to :

(A) $\sim p \wedge q$ (B) $p \wedge q$ (C) $p \vee q$ (D) $p \vee \sim q$

Q111. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The negation of $\sim s \vee (\sim r \wedge s)$ is equivalent to :

(A) $s \vee (r \vee \sim s)$ (B) $s \wedge r$
(C) $s \wedge \sim r$ (D) $s \wedge (r \wedge \sim s)$

Q112. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

The statement $\sim (p \leftrightarrow \sim q)$ is :

(A) equivalent to $\sim p \leftrightarrow q$ (B) a tautology
(C) a fallacy (D) equivalent to $p \leftrightarrow q$

Q113. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

Consider :
Statement – I : $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is a fallacy.
Statement – II : $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is a tautology.

(A) Statement - I is True; Statement -II is true; Statement-II is not a correct explanation for Statement-I
(B) Statement -I is True; Statement -II is False.
(C) Statement -I is False; Statement -II is True
(D) Statement -I is True; Statement -II is True; Statement-II is a correct explanation for Statement-I

Q114. AIEEE 2012 · MCQ (Single Correct Answer)

The negation of the statement "If I become a teacher, then I will open a school" is :

(A) I will become a teacher and I will not open a school

(B) Either I will not become a teacher or I will not open a school

(C) Neither I will become a teacher nor I will open a school

(D) I will not become a teacher or I will open a school

Q115. AIEEE 2011 · MCQ (Single Correct Answer)

Consider the following statements
P : Suman is brilliant
Q : Suman is rich
R : Suman is honest
The negation of the statement,

"Suman is brilliant and dishonest if and only if Suman is rich" can be expressed as :

(A) $\sim [Q \leftrightarrow (P \wedge \sim R)]$

(B) $\sim Q \leftrightarrow P \wedge R$

(C) $\sim (P \wedge \sim R) \leftrightarrow Q$

(D) $\sim P \wedge (Q \leftrightarrow \sim R)$

Q116. AIEEE 2010 · MCQ (Single Correct Answer)

Let S be a non-empty subset of R. Consider the following statement:
P : There is a rational number x ∈ S such that x > 0.
Which of the following statements is the negation of the statement P?

(A) There is no rational number x ∈ S such that x ≤ 0

(B) Every rational number x ∈ S satisfies x ≤ 0

(C) x ∈ S and x ≤ 0 $\Rightarrow$ x is not rational

(D) There is a rational number x ∈ S such that x ≤ 0

Q117. AIEEE 2009 · MCQ (Single Correct Answer)

Statement-1 : $\sim (p \leftrightarrow \sim q)$ is equivalent to $p \leftrightarrow q$.
Statement-2 : $\sim (p \leftrightarrow \sim q)$ is a tautology.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1

(C) Statement-1 is true, Statement-2 is false

(D) Statement-1 is false, Statement-2 is true

Q118. AIEEE 2008 · MCQ (Single Correct Answer)

Let p be the statement “x is an irrational number”, q be the statement “y is a transcendental number”, and r be the statement “x is a rational number iff y is a transcendental number”.

Statement –1: r is equivalent to either q or p.

Statement –2: r is equivalent to $\sim (p \leftrightarrow \sim q)$

(A) Statement – 1 is false, Statement – 2 is false

(B) Statement –1 is false, Statement –2 is true

(C) Statement –1 is true, Statement –2 is true, Statement –2 is a correct explanation for Statement –1

(D) Statement –1 is true, Statement –2 is true; Statement –2 is not a correct explanation for Statement –1

Q119. AIEEE 2008 · MCQ (Single Correct Answer)

The statement $p \rightarrow (q \rightarrow p)$ is equivalent to

(A) $p \rightarrow (p \leftrightarrow q)$

(B) $p \rightarrow (p \rightarrow q)$

(C) $p \rightarrow (p \vee q)$

(D) $p \rightarrow (p \wedge q)$

Q120. JEE Main 2023 (Online) 11th April Morning Shift · Numerical

The number of ordered triplets of the truth values of p, q and r such that the truth value of the statement $(p \vee q) \wedge (p \vee r) \Rightarrow (q \vee r)$ is True, is equal to _____.

Q121. JEE Main 2022 (Online) 28th June Evening Shift · Numerical

The maximum number of compound propositions, out of p∨r∨s, p∨r∨∼s, p∨∼q∨s, ∼p∨∼r∨s, ∼p∨∼r∨∼s, ∼p∨q∨∼s, q∨r∨∼s, q∨∼r∨∼s, ∼p∨∼q∨∼s that can be made simultaneously true by an assignment of the truth values to p, q, r and s, is equal to _____.

Q122. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (More than One Correct Answer)

The statement $B \Rightarrow ((\sim A) \vee B)$ is equivalent to :

(A) $B \Rightarrow ((\sim A) \Rightarrow B)$

(B) $A \Rightarrow (A \Leftrightarrow B)$

(C) $A \Rightarrow ((\sim A) \Rightarrow B)$

(D) $B \Rightarrow (A \Rightarrow B)$

Answer Key

1. **B**	2. **A**	3. **B**	4. **C**	5. **C**
6. **C**	7. **A**	8. **A**	9. **B**	10. **A**
11. **D**	12. **D**	13. **A**	14. **A**	15. **C**
16. **C**	17. **B**	18. **A**	19. **A**	20. **A**
21. **A**	22. **C**	23. **B**	24. **D**	25. **D**
26. **B**	27. **D**	28. **C**	29. **D**	30. **D**
31. **D**	32. **D**	33. **A**	34. **C**	35. **C**
36. **C**	37. **C**	38. **D**	39. **C**	40. **A**
41. **C**	42. **C**	43. **B**	44. **B**	45. **D**
46. **A**	47. **C**	48. **A**	49. **A**	50. **C**
51. **C**	52. **A**	53. **D**	54. **C**	55. **D**
56. **D**	57. **B**	58. **B**	59. **D**	60. **B**
61. **A**	62. **C**	63. **C**	64. **A**	65. **B**
66. **A**	67. **C**	68. **B**	69. **D**	70. **D**
71. **A**	72. **A**	73. **C**	74. **A**	75. **D**
76. **A**	77. **A**	78. **C**	79. **A**	80. **A**
81. **B**	82. **D**	83. **A**	84. **B**	85. **A**
86. **C**	87. **D**	88. **B**	89. **B**	90. **B**
91. **C**	92. **A**	93. **B**	94. **D**	95. **B**
96. **D**	97. **B**	98. **B**	99. **C**	100. **C**
101. **D**	102. **B**	103. **A**	104. **B**	105. **A**
106. **B**	107. **D**	108. **D**	109. **C**	110. **C**
111. **B**	112. **D**	113. **A**	114. **A**	115. **A**
116. **B**	117. **C**	118. **A**	119. **C**	120. **7**
121. **9**	122. **A, C, D**			